Tissue specificity of Isoforms across Species

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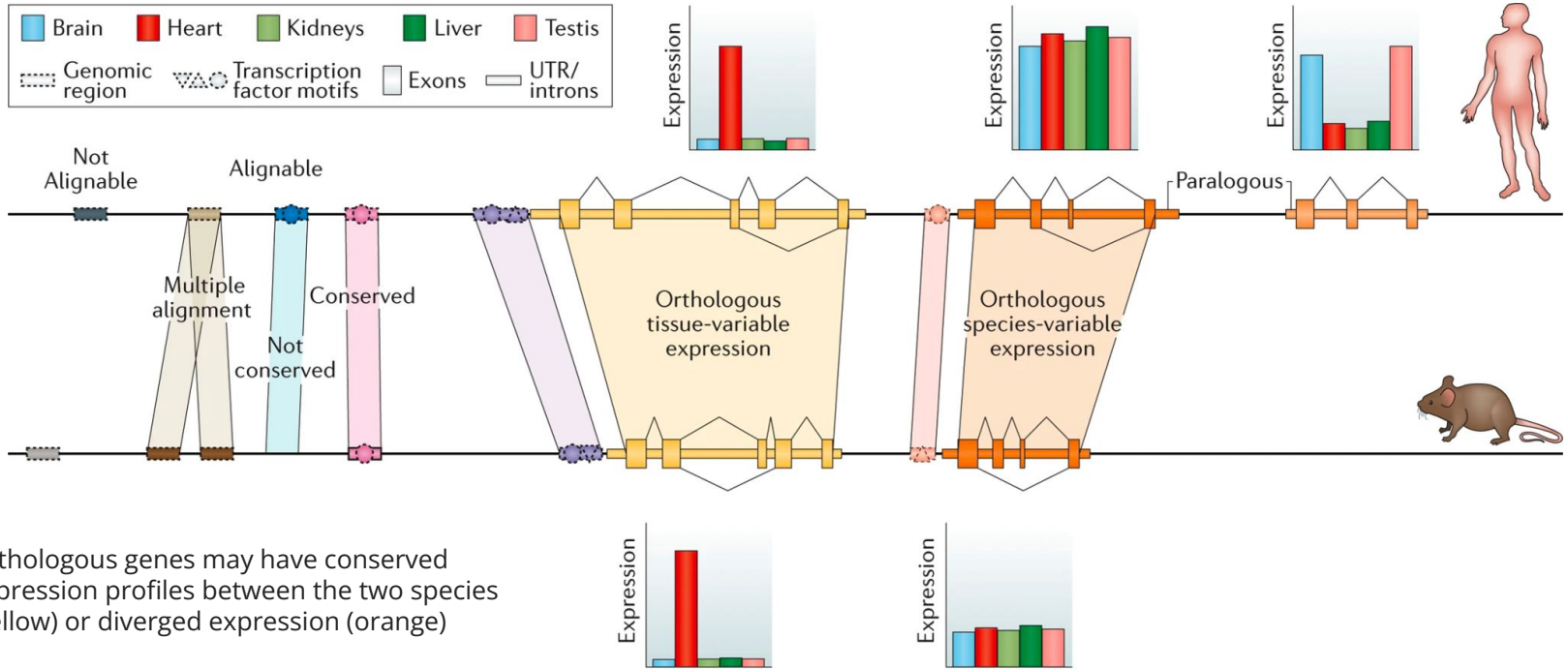
Motivation

- We wanted to analyze the conservation across species.
- Why mouse?



Orthologous genes may have conserved expression profiles between the two species (yellow) or diverged expression (orange)

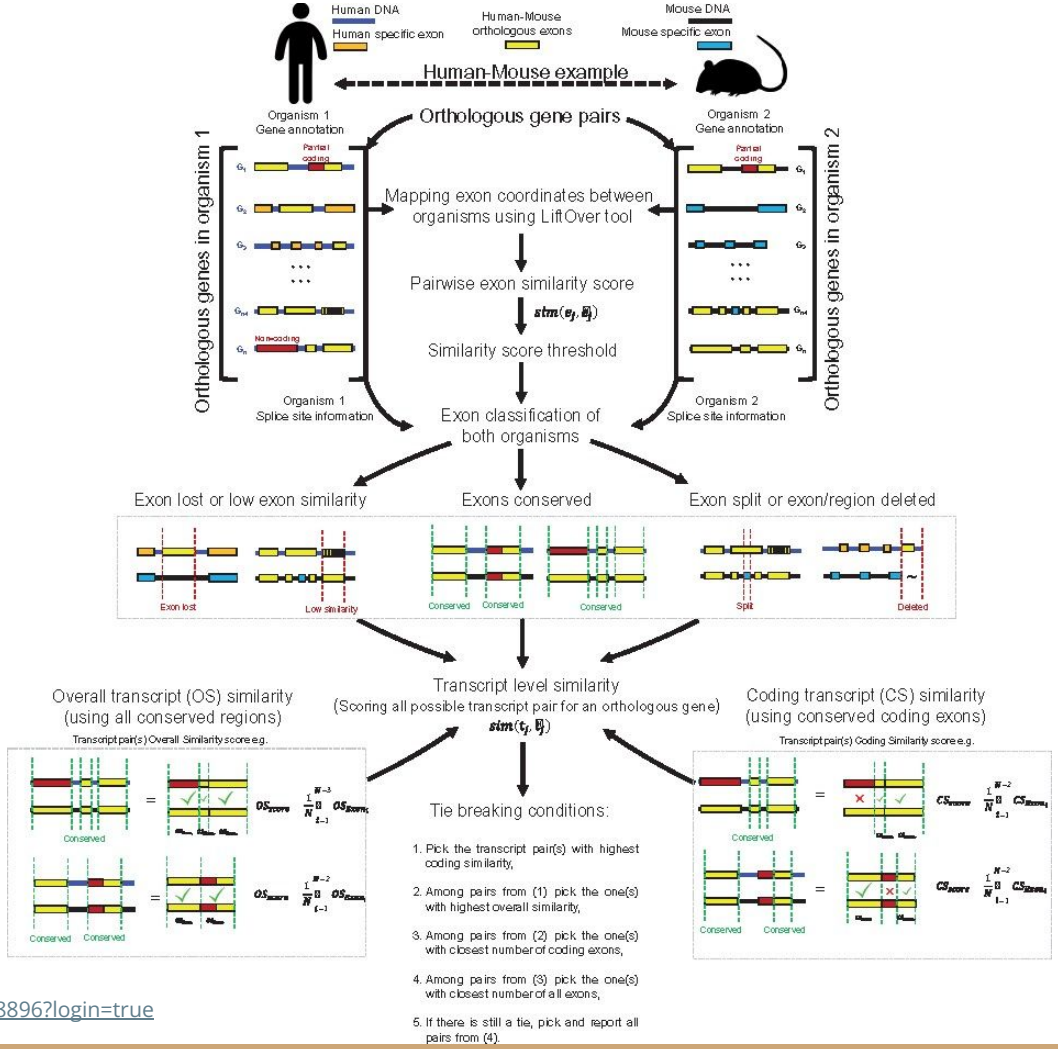
Ref: https://www.nature.com/articles/nrg.2017.19
Ref: https://www.genome.gov/10001345/importance-of-mouse-genome

Nature Reviews | Genetics

Motivation for the method ExTraMapper

- Limited number of published studies that aim to find exon-level relationships among different organisms
- Most methods consider only coding exons.
- Limitations of other methods like Exalign:
  - The method does not utilize any sequence similarity or orthology information between the exons or the transcripts
  - Bias towards assigning higher similarity scores for transcripts with large number of exons

Pipeline



Ref: https://academic.oup.com/bioinformatics/article/37/20/3412/6278896?login=true

Questions we now want to answer

- Which protein domains are conserved between human and mouse?
- Which are disrupted in alternative splicing? (some domains are lost or gained by splicing)
- Applying methods (AI) to compare one cancer v/s another. Status of conserved domains in cancer?
- Tissue specific expression of those conserved across species in different conditions (normal v/s cancer) for different tissues

Other questions of interest

- Drug target interactions:
  - Between multiple sequences - based on conservation scores.
  - Then apply those on human and mouse and summarize the expression patterns based on 3D molecular structures

Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6954184/

To Do

- Understand alternative transcription and Understand conservation of genes
- Read ExtraMapper - Aim is to understand one-to-one orthology mapping between human and mouse at exon and transcript level. Why is this study important?
- Identify unique (only present in one organism) alternatively spliced exons and corresponding transcripts in each organism
    - Human v/s chimpanzee,

To Do Cont...

- Explore the expression patterns of each set of transcripts (Normal tissues)

    1. orthologous transcripts,

    2. Unique transcripts

- Explore the expression of 1 and 2 above in GTeX (blood, brain and lung) and their corresponding cancers (TCGA) of:
    - tissue specific transcripts
    - Differentially expressed transcripts in tumors

To Do Cont...

1. Set of first exons conserved between human and mouse and see if DNABERT can classify those.
2. Misclassification of LUAD and LUSC by homogeneous grouping of tumors (Subtyping idea)

Readings

- Alternative promoter usage in Mammalian genomes - https://www.clinicalkey.com/service/content/pdf/watermarked/1-s2.0-S0168952508000590.pdf?locale=en_US&searchIndex=

- ExTraMapper Paper - https://academic.oup.com/bioinformatics/article/37/20/3412/6278896?login=true
- Code - https://github.com/ay-lab/ExTraMapper
- Orthoexon paper - https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3303729/

- NCBI resources:
  - Conserved Domain Database https://www.ncbi.nlm.nih.gov/cdd
  - Homologene https://www.ncbi.nlm.nih.gov/homologene